



Cryptographic Hashing

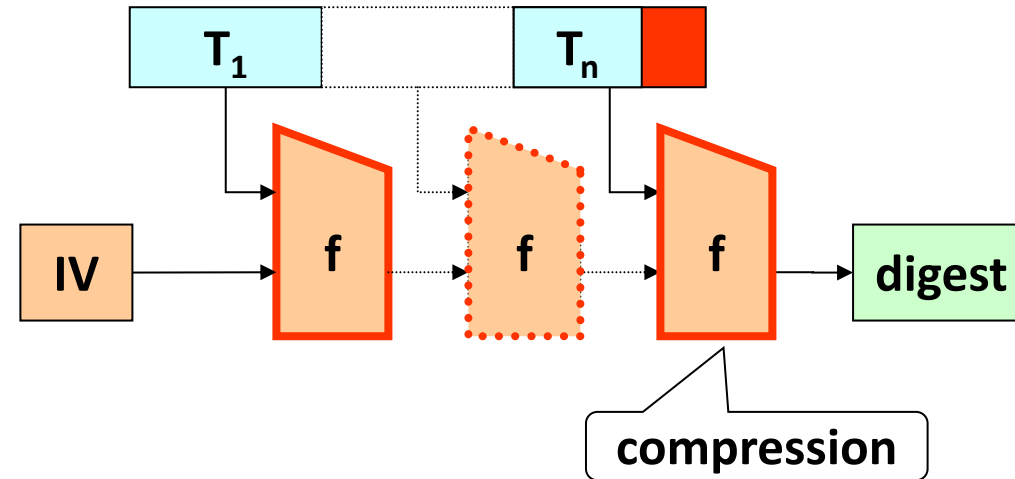
(digest functions)



Digest functions

- Give a fixed-length value from a variable-length text
 - Sort of text “fingerprint”
- Produce very different values for similar texts
 - Cryptographic one-way hash functions
- Relevant properties:
 - Preimage resistance
 - Given a digest, it is infeasible to find an original text producing it
 - 2nd-preimage resistance
 - Given a text, it is infeasible to find another one with the same digest
 - Collision resistance
 - It is infeasible to find any two texts with the same digest
 - Birthday paradox

Digest functions: approaches



- Merkle-Damgård construction
 - Iterative compression
 - Collision-resistant, one-way compression functions
 - Length padding (1, followed by zeros, followed by length)
- Sponge functions
 - **Absorption**: update a finite internal state (entropy pool) from a variable-length, padded input stream
 - **Squeezing**: produce an arbitrary-length output from the internal state

Digest functions: common algorithms

- MD₄ (128 bits)
 - Still used, but very easy to break
- MD₅ (128 bits)
 - Very easy to find collisions!
 - Disclaimer: it can be used when collisions are not an issue
- SHA-1 (Secure Hash Algorithm, 160 bits)
 - Also no longer secure ... (collisions found in 2017)
- RIPEMD (128 and 160)
- HAVAL (128, 160, 192, 224, 256)
- SHA-2 family (SHA-256, SHA-384, SHA-512)
- SHA-3 family (SHA3-224, SHA3-256, SHA3-384, SHA3-512)
- Blake2s (128, 160, 192, 224, 256)
- Blake2b (160, 256, 384, 512)

Digest functions: what are they good for?

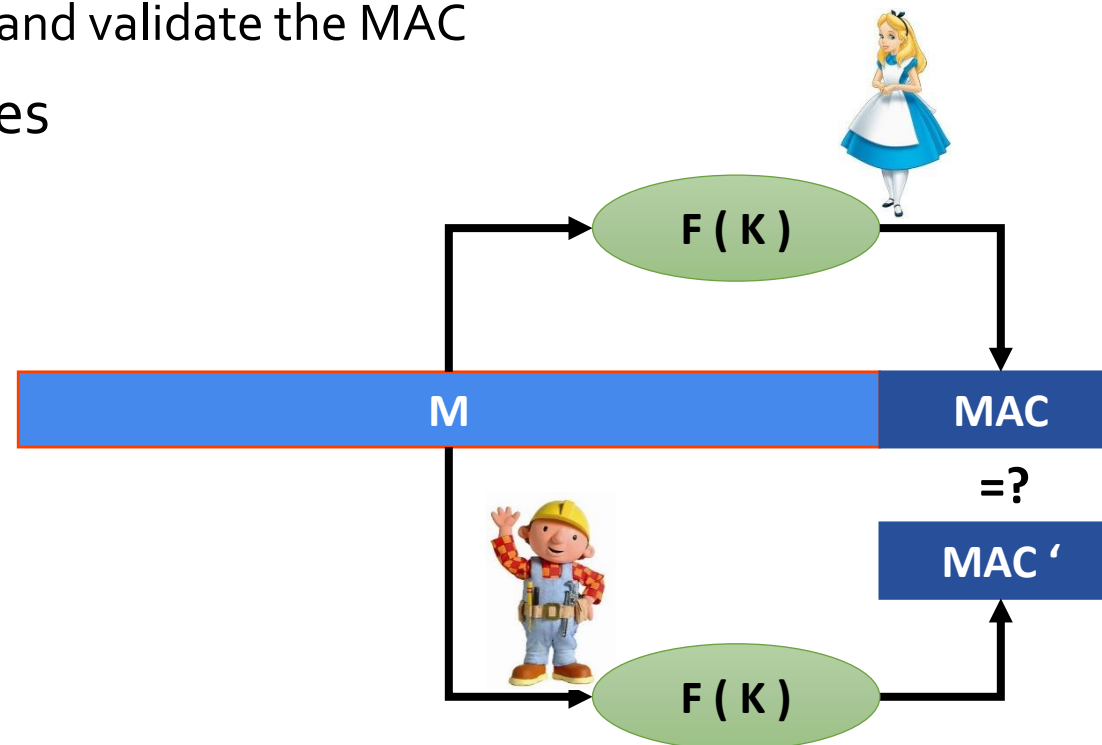
- To produce robust or secure checksums
 - Message Integrity Codes (MIC)
 - Message Authentication Codes (MAC)
- To produce fingerprints of documents
 - Acts as a document “identity”
 - Useful for signing purposes
- To implement key derivation processes
 - Password → key
 - Key → key
- To implement cryptopuzzles
 - E.g. Bitcoin mining

Message Integrity Code (MIC)

- Provide the capability to detect changes by devices
 - Communication/storage errors
 - From a random process or without control
- Send: Calculate MIC and send $T + MIC$
 - $T = \text{Text}$
 - $MIC = \text{digest}(T)$
- Receive: Receive data (T') and check if $H(T) = MIC$
 - Calculate $MIC' = \text{digest}(T')$
 - Validate if $MIC' = MIC$
- Does not protect from planned changes to the text
 - Attacker can manipulate T into T'' and calculate a new MIC''

Message Authentication Codes (MAC)

- Hash, or digest, computed with a key
 - Only key holders can generate and validate the MAC
- Used to authenticate messages
 - $M' = M \mid \text{MAC}(M)$



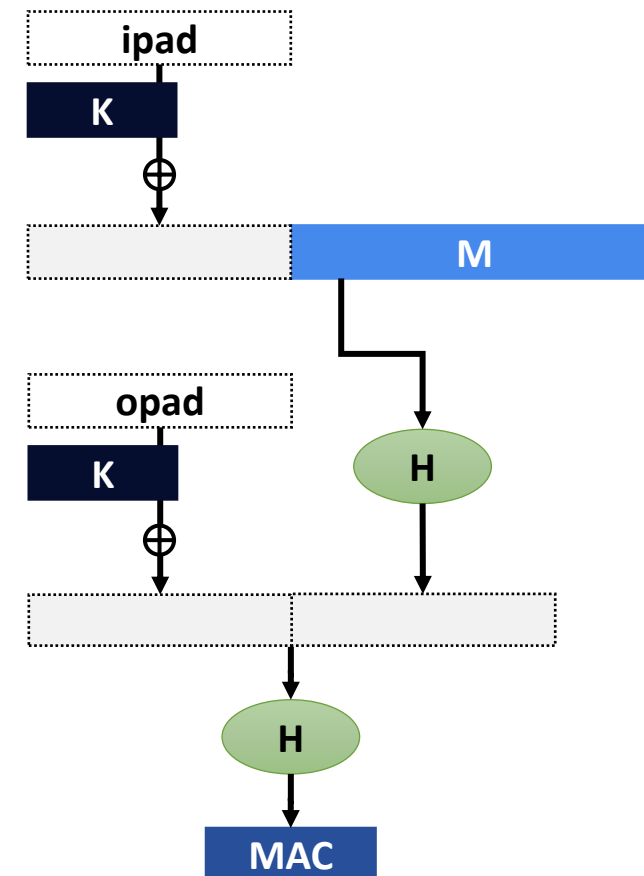
MAC functions: Hash-based approaches

- Adding a key to the hashed data
 - Keyed-MD5 (128 bits)
 - $MD5(K, \text{keyfill}, \text{text}, K, MD5\text{fill})$
 - HMAC (Hashed-based MAC)
 - Generic construction, uses a hash function H
 - Output length depends on H
 - HMAC-MD5, HMAC-SHA, etc.

$H(K, \text{opad}, H(K, \text{ipad}, \text{text}))$

$\text{ipad} = \text{0x36 B times}$

$\text{opad} = \text{0x5C B times}$



Recommended Reading / Resources

- Security in Computing, 4th edition, C. P. Pfleeger, S. L. Pfleeger: Chaps. 2 & 3
- Segurança em Redes Informáticas, A. Zúquete: Chap. 2

Additional Resources:

- [Digest function \(cryptographic hash function\)](#)
- [Birthday paradox](#)